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To cite this article: Magdalena Zawadka, Jakub Smolka, Maria Skublewska-Paszkowska, Edyta Lukasik, Mirosław Jablonski & Piotr Gawda (2022): The influence of sedentary behaviour on lumbar-pelvic kinematics during squatting and forward bending among physically active students, *Ergonomics*, DOI: [10.1080/00140139.2022.2061051](https://doi.org/10.1080/00140139.2022.2061051)

To link to this article: <https://doi.org/10.1080/00140139.2022.2061051>



Published online: 15 Apr 2022.



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ARTICLE



The influence of sedentary behaviour on lumbar-pelvic kinematics during squatting and forward bending among physically active students

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ABSTRACT

Prolonged sitting may involve several mechanisms that make it a risk factor for low back pain. The aim of this study was to investigate lumbar-pelvic kinematics and multifidus muscle (MF) activity during squatting and forward bending in relation to the sedentary behaviour of physically active students. Sixty-three students were divided into two groups according to the time spent in a sitting position during the day: 'high' (>7 h/day); 'low' (≤7 h/day). Lumbar-pelvic ratios, ranges of motion, angular velocities, and MF flexion-relaxation phenomenon were investigated. Data were obtained using the optical motion analysis system, and surface electromyography. The results indicated that lumbar-pelvic ratios during both tasks and velocity of lumbar spine during squatting were significantly greater in the 'high' than in the 'low' sitting group. Muscle activity showed no differences between groups. Prolonged sitting can be considered a factor that slightly, but statistically significantly influences the lumbar-pelvic kinematics in physically active people.

Practitioner summary: Lumbar-pelvic kinematics can be altered by prolonged sitting in physically active students. Lumbar-pelvic ratios during squatting and forward bending and lumbar spine velocity during squatting were significantly greater in the 'high' than in the 'low' sitting group. Sedentary behaviour should be considered during an assessment of movement patterns.

Abbreviations: BMI: body mass index; ERR: extension-relaxation ratio; FRP: flexion-relaxation phenomenon; FRR: flexion-relaxation ratio; IPAQ: International Physical Activity Questionnaire; LBP: low back pain; METs: metabolic equivalent of tasks; MF: multifidus muscle; PA: physical activity; ROM: range of motion; sEMG: surface electromyography

ARTICLE HISTORY

Received 18 August 2021
Accepted 23 March 2022

KEYWORDS

Physical activity;
electromyography; sitting
position; spine

1. Introduction

Sedentary behaviour and physical activity (PA) are considered as determinant factors of health and life span (Stamatakis and Gill 2019; Chau et al. 2013). Sitting time can elevate cardiovascular disease risk even in those who are highly physically active, and PA does not eliminate the increased risk of death associated with high sitting time (Ekelund et al. 2016; Stamatakis Emmanuel et al. 2019). Sedentary behaviour is also related to low back pain (LBP) (Hanna et al. 2019). Previous research on sitting posture and muscle activation suggests that habitually sustaining flexed sitting postures for long periods of time may result in the deconditioning of spine muscles and lead to a greater risk of back injury or a provocation of existing

LBP (Bontrup et al. 2019; Dankaerts et al. 2006; P. O'Sullivan et al. 2006; P. B. O'Sullivan et al. 2006).

Sedentary behaviour is defined as any waking behaviour that is characterised by an energy expenditure of 1.5 metabolic equivalent of tasks (METs), while in a sitting, reclining, or lying posture (Gardner et al. 2016). Self-report sitting behaviours were previously defined in the literature as sedentary when the total time of sitting (transport, work leisure activity) during a day exceeded 7 h (Heneghan et al. 2018; van Uffelen et al. 2013; Park et al. 2018). Sufficient PA, according to the International Physical Activity Questionnaire (IPAQ), is characterised by 5 or more days of any combination of walking, moderate intensity or vigorous intensity achieving a minimum total PA of at least 600 MET minutes a week (Biernat 2013).

The increased time spent in sitting is a global problem and office work is a major contributor to sedentary behaviour (Kazi et al. 2019). However, long sitting is also particularly characteristic of young people continuing their education (Matusiak-Wieczorek et al. 2020). Students are one of the most sedative groups in European populations (Loyen et al. 2016; Jelsma et al. 2019). With interest in a healthy life style growing among young adults, more and more of them meet the PA recommendation (Matusiak-Wieczorek et al. 2020). Therefore, high PA does not exclude sitting behaviours, which are associated not only with education but also with leisure activities like watching television, playing computer games or reading. At the same time one person can be considered physically active and spend a lot of time sitting. It seems unclear how long sitting times can influence persons who meet PA guidelines. Higher amounts of daily total sitting time have been shown to be associated with greater risk of all-cause mortality and moderate or vigorous physical activity appears to attenuate this hazardous association (Chau et al. 2013).

During trunk flexion postures the lumbodorsal fascia and spinal ligaments provide necessary support for the extensor muscles, reducing the necessity of contraction to maintain the fully flexed posture. Observation of electrical silence of the back extensors muscles at full trunk flexion is defined as flexion-relaxation phenomenon and calculated as flexion-relaxation ratio and/or extension-relaxation ratio (McGorry and Lin 2012; Laird et al. 2019). During flexed postures the transition of load moves towards other active and/or passive structures close to the end range of flexion (P. O'Sullivan et al. 2006). A seated posture can be affected by the parameters of a chair, but in general a sitting posture causes the pelvis to rotate backward, reduce lumbar lordosis and increase intervertebral disc pressure (Harrison et al. 1999). Long-duration sitting can be a contributory factor in hamstring tightness (Waqas et al. 2016; Fatima et al. 2017). Reduced hamstrings flexibility has been reported as a factor altering lumbar-pelvic mobility during functional tasks like forward bending or reaching tasks (Gajdosik, Hatcher, and Whitsell 1992; Zawadka et al. 2018). Previously lumbar-pelvic motion was widely examined to demonstrate the relationship between LBP and lumbar-hip coordination during many activities (Larivière, Gagnon, and Loisel 2000; Esola et al. 1996; Shojaei, Salt, et al. 2017). One of the parameters described in the literature is the lumbar-pelvic ratio defined as the ratio of total lumbar motion to pelvis motion in the sagittal plane (Zawadka et al. 2018; Kasahara et al. 2008). Lumbar-pelvic motion has

been shown to be connected to LBP by alternations in lumbar-hip coordination, slower task pace, decrease in lumbar contribution in overall movement and a compensatory increase in mobility of other spine segments and pelvis (Larivière, Gagnon, and Loisel 2000; Esola et al. 1996; Shojaei, Salt, et al. 2017). Changes in the trunk flexion-extension movement pattern may be the result of a neuromuscular adaptation to minimise deformation in the lower back passive tissues and avoid pain (Shojaei, Salt, et al. 2017). However, a cause-and-effect relationship has not been well established so far. Due to the high functional character of movement, the biomechanics of the spine and pelvis during squatting and forward bending were often studied (Zawadka et al. 2020; Shojaei, Vazirian, et al. 2017). However, the relationship between sitting behaviour and lumbar kinematics in standing among students is not well documented in the literature.

Due to the fact that prolonged sitting involve several mechanisms that make it a risk factor for LBP (Park et al. 2018) and that LBP is clearly related to biomechanical alternations during trunk flexion-extension tasks, we decided to investigate the linkage between sitting time and lumbar-pelvic kinematics. Understanding lumbar-pelvic kinematics in an asymptomatic group of long-sitting young people could provide useful information about the missing link between sitting behaviour and biomechanical changes during dynamic activities observed in LBP patients.

Until recently biomechanical alternations of dynamic activities as a consequence of prolonged sitting posture have not been widely investigated. Understanding the influence of prolonged sitting on lumbar-pelvic coordination during trunk flexion in dynamic tasks has important clinical implications in assessing and preventing LBP among students. Therefore, the hypothesis was that an increased time of sitting would alter lumbar-pelvic kinematics and muscle activity during squatting and forward bending compared to less sitting time. The aim of this study was to investigate lumbar-pelvic kinematics and multifidus muscle (MF) activity during squatting and forward bending in relation to the sedentary behaviour of physically active university students.

2. Materials and methods

2.1. Participants

Sixty three students participated in this study: 23 females and 40 males. Participants were excluded if they had any musculoskeletal pain or an orthopaedic injury in their history or during the examination. The International Physical Activity Questionnaire long

version was used for calculating participants' PA and sitting times. Subjects met the criteria of high or moderate physical activity level. The PA was measured in MET units (energy expenditure as reflected by oxygen consumption) (Lear et al. 2017). The average sitting time included sitting during weekdays, weekend and in moving vehicles.

Participants were divided into groups according to their sedentary behaviour (time spent in a sitting position during day): high (>7 h/day); low (≤ 7 h/day) (Heneghan et al. 2018; Park et al. 2018). There were 29 participants (9 females and 20 males) in the 'high' sitting group and 34 participants (14 females and 20 males) in the 'low' sitting group. The numbers of females and males in both groups were not statistically different (chi squared = 0.69; $p = 0.40$). The study was approved by the Ethical Committee at the Medical University of Lublin (KE-0254/322/2018) and conducted according to the Declaration of Helsinki. Tables 1 and 2 show comparisons between groups and between sexes according to age, PA, sitting time, body mass, height and Body Mass Index (BMI). There were no statistically significant differences in the above mentioned variables between 'low' and 'high' sitters except the sitting time (Me = 5.32 h/day vs Me = 9.14 h/day; $F = 145.87$, $p < 0.001$, $\eta_p^2 = 0.71$). However, there were differences between females and males in body mass ($F = 41.50$, $p < 0.001$, $\eta_p^2 = 0.41$) and height ($F = 89.17$, $p < 0.001$, $\eta_p^2 = 0.60$), resulting from anatomical dimorphism. Females were also more physically active than men ($F = 5.03$, $p = 0.03$, $\eta_p^2 = 0.08$), particularly in domestic activities ($F = 10.46$, $p < 0.01$, $\eta_p^2 = 0.15$), but there were no differences in sitting time between males and females (Me = 6.48 h/day vs Me = 7.11 h/day; $F = 0.25$, $p = 0.61$).

Table 1. Age, body mass, height and BMI of participants.

Variables	Low sitting time			High sitting time			Females			Males		
	M	Me	SD	M	Me	SD	M	Me	SD	M	Me	SD
Age [years]	20.71	20.00	1.14	20.79	21.00	0.82	20.30**	20.00	0.47	21.00**	21.00	1.13
Body mass [kg]	70.12	67.75	12.92	70.84	70.00	14.45	59.31***	57.40	7.14	76.86***	75.30	12.14
Height [m]	1.75	1.75	0.08	1.75	1.80	0.11	1.66***	1.66	0.06	1.81***	1.80	0.06
BMI [kg/m ²]	22.65	22.48	2.85	22.89	22.31	2.98	21.42**	20.83	1.87	23.53**	23.59	3.10

M: Mean; Me: median; SD: standard deviation; ** $p < 0.01$, *** $p < 0.001$.

Table 2. Physical activity (PA) and sitting time of participants.

PA [METmin/week]	Low sitting time			High sitting time			Females			Males		
	M	Me	SD	M	Me	SD	M	Me	SD	M	Me	SD
Studying	1174.38	60.00	1810.61	839.67	148.50	1275.80	1276.48	160.00	2005.98	873.01	0.00	1287.10
Walking from place to place	1290.62	774.00	1193.44	1520.90	1386.00	1032.99	1590.59	1386.00	1216.36	1285.09	823.50	1059.72
Domestic and garden activities	728.38	450.00	852.12	934.66	630.00	974.74	1222.39**	870.00	1111.33	593.88**	382.50	684.83
Leisure activities (sport and recreation)	1858.52	1521.75	1502.20	1952.38	1455.00	1861.32	1999.94	1546.00	1584.28	1845.25	1447.50	1724.96
Total PA	5037.59	4790.20	2456.34	5256.53	4989.00	3074.84	6059.74*	6354.00	2536.08	4608.59*	4401.00	2737.86
Sitting time [hours/day]	5.01***	5.32	1.35	9.34***	9.14	1.41	6.55	6.48	2.90	7.27	7.11	2.36

M: Mean; Me: median; SD: standard deviation; * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

2.2. Procedures and measurements

Two tasks were investigated in this study: forward bending and squatting. Participants performed five trials of each task without additional load. They were not trained for these tasks prior to the investigation. Before recording the trials participants were instructed in how to do the tasks, and the execution of tasks was demonstrated to them. The initial position of both tasks was the same: standing looking straight ahead with arms forward, parallel to the floor and with feet at shoulder's width apart. Forward bending was executed to the maximal depth with straightened knees, and squatting was executed to the maximal depth maintaining heel contact throughout the task. The pace of movement was self-selected; however, participants were instructed not to do any unpredictable or sudden movements. Three seconds after reaching the final position of forward bending/squatting, participants were asked to return to their initial positions at a similar pace. An average of three middle trials (without the first and the last) was statistically analysed.

Kinematic data were obtained using the VICON motion analysis system (Oxford Metrics Ltd, Oxford, UK) with 8 infra-red cameras operating at a sampling rate of 100 Hz. A set of reflective markers was placed on specific anatomical landmarks. Markers have been attached to the following locations to build the spine and pelvis model: tenth and twelfth thoracic spinous process (Th 10, Th 12), posterior and anterior superior iliac spine and second and third sacral spinous process (S2, S3). The system registered three-dimensional trajectories of passive markers. Markers trajectories were filtered and segments of the lumbar spine and pelvis were modelled using software (Vicon, Nexus; Oxford Metrics).

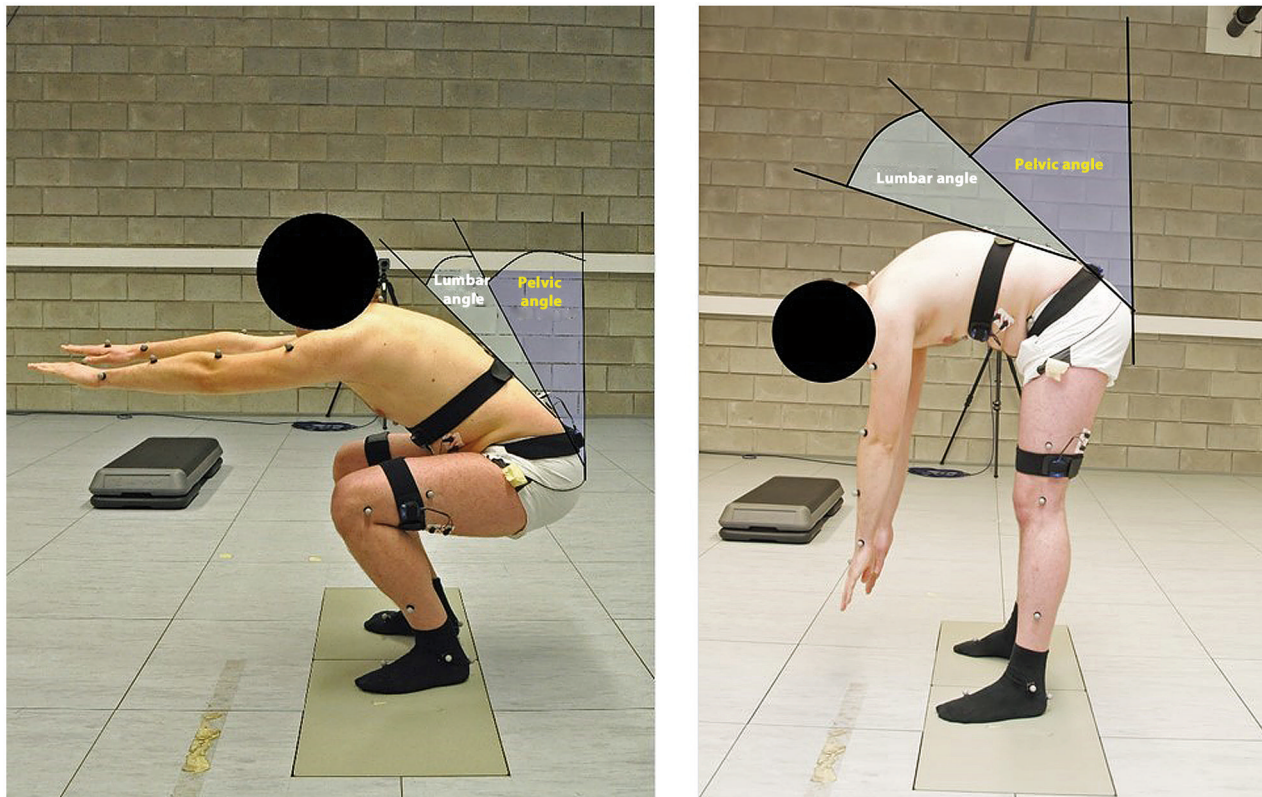


Figure 1. Testing postures and angles of the pelvis and lumbar spine.

The pelvis angle was defined as the angle of the sacral bone relative to the vertical line, and the lumbar spine angle was defined as the angle of the thoracic spine relative to the sacral bone. The angles were computed in the sagittal plane (Figure 1). The range of motion (ROM) was defined as the difference between the maximal and minimal angle obtained by each segment in the sagittal plane during a task. The lumbar-pelvic ratio describes the coordination during spine flexion-extension tasks (also called the lumbar-pelvic rhythm) (S. Shin and Yoo 2020). In this study ratios were calculated by dividing the total lumbar ROM by the pelvis ROM. Angular velocity of segments (spine, pelvis) in sagittal plane over the change in time. Average and peak velocities were analysed during two phases of movement: descent and ascent.

Surface electromyography (sEMG) integrated with the motion capture system (wireless EMG; Myon) was used to record superficial lumbar multifidus muscle activity at the sampling rate of 1000 Hz. Electrodes were placed on the back along the axes of the muscle fibres, according to the SENIAM protocol: on and aligned with a line from caudal tip of posterior superior iliac spine to the interspace between L1 and L2 interspace at the level of L5 spinous process (i.e. about

2–3 cm from the midline) ('Sensor Locations' 2021). Two pairs of self-adhesive disposable Ag/AgCl disc surface electrodes (SORIMEX, Toruń, Poland) with an electrical contact surface of 1 cm² were placed bilaterally, with a 2 cm interelectrode distance. The subjects' hair was shaved when it was necessary and the skin was cleaned with alcohol before the sEMG electrode was affixed.

The EMG signals collected during the movements were band-pass filtered at 10–450 Hz using a fourth-order Butterworth filter with zero lag and then rectified. EMG data were smoothed using a 100 millisecond moving average window. Based on the methodology of previous studies, the peak amplitude of EMG was determined within the ascent and descent phases, and the mean for a 200 ms window about the peak was taken as the sEMG amplitude value (McGorry and Lin 2012; Neblett et al. 2003). A flexion-relaxation ratio (FRR) and extension-relaxation ratio (ERR) of the activity during the forward bending and full squat was calculated to determinate the presence of the flexion-relaxation phenomenon (FRP) and muscle silence (McGorry and Lin 2012). For this purpose, we have divided tasks into the descent phase, the static position phase and the ascent phase. The value for the FRR was calculated by dividing the mean of a 200 ms window about the peak

amplitude of descent phase (1) and the mean of a 200 ms about midpoint of static position phase (2). The ERR was calculated by dividing the mean of a 200 ms window about the peak amplitude of ascent phase (1) and the mean of a 200 ms midpoint of static position phase. Higher ratios indicate relatively more flexion-relaxation phenomenon (less activation) of the lumbar multifidus muscle.

2.3. Statistics

Statistical analyses were conducted using the Statistica software (ver. 13.1 TIBCO Inc., Palo Alto, CA, USA). Continuous data were described in tables by mean (M), median (Me) and standard deviation (SD). Box and whisker plots show median, upper and lower quartiles and maximum and minimum values. The level of statistical significance was set at $p=0.05$. Normal distribution of data was verified using the Shapiro-Wilk test, and the nonparametric statistical method was used for kinematic and EMG analyses. The Chi-square test was used for comparing the number of females and males in groups and two-way ANOVA with partial eta squared (η_p^2) was used for comparing group characteristics (Tables 1 and 2). In order to compare lumbar-pelvic kinematics between 'low' and 'high' sitting groups and males and females we used the Mann-Whitney U-test. The intra-class correlation coefficients (ICC) show good to excellent reliability of the kinematic data (ICC = 0.90–0.97 for squatting and ICC = 0.78–0.97 for forward bending). The Glass rank-biserial correlation was used to calculate effect sizes (ES) of statistically significant comparisons, and they were interpreted as small (0.2), medium (0.5),

and large (0.8) effect sizes (Tomczak and Tomczak 2014; Fritz, Morris, and Richler 2012).

3. Results

3.1. Lumbar-pelvic ratios and ROMs

As shown in Figure 2, the lumbar-pelvic ratio was significantly greater in the 'high' than in the 'low' sitting group (Median (Me) = 2.37 vs Me = 1.31; $Z = -2.53$, $p = 0.01$, ES = 0.37) during squatting. The lumbar-pelvic ratio during forward bending was also significantly greater in the 'high' than in 'low' sitting group (Me = 2.29 vs Me = 1.73; $Z = -2.02$, $p = 0.04$, ES = 0.30). However, there were no statistically significant differences between groups in pelvic ROM (Me = 17.39° vs Me = 20.93°; $Z = 1.81$, $p = 0.07$) and spine ROM (Me = 38.98° vs Me = 32.88°; $Z = 1.43$, $p = 0.15$) during squatting (Figure 3). Spine ROM was significantly greater in the 'high' than in the 'low' sitting group during forward bending (Me = 75.69° vs Me = 68.94°; $Z = -2.31$, $p = 0.02$, ES = 0.34). There were no statistically significant differences between groups in pelvic ROM during forward bending (Me = 36.48° vs Me = 39.96°; $Z = 1.33$, $p = 0.18$) (Figure 4).

Males showed greater lumbar-pelvic ratio than females in both, squatting (Me = 2.34 vs Me = 1.16; $Z = -3.36$, $p < 0.001$, ES = 0.51) and forward bending (Me = 2.24 vs Me = 1.51; $Z = -3.30$, $p < 0.001$, ES = 0.50). Lumbar ROM was significantly greater in males than in females during squatting (Me = 39.38 vs Me = 26.88; $Z = 3.03$, $p < 0.01$, ES = 0.46). During forward bending males showed greater lumbar ROM

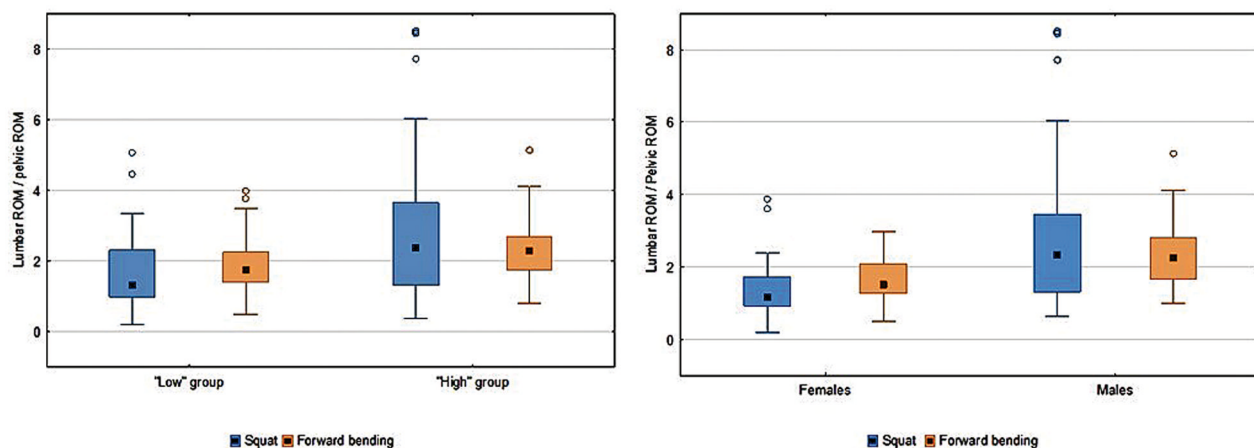


Figure 2. Ratios of the lumbar and pelvic ranges of motion (ROM) during squatting and forward bending. Boxes show median, upper and lower quartiles. Whiskers show maximum and minimum (1.5 of interquartile range). Dots show outliers (above 1.5 of interquartile range). All comparisons between groups and sexes are statistically significant ($p < 0.05$).

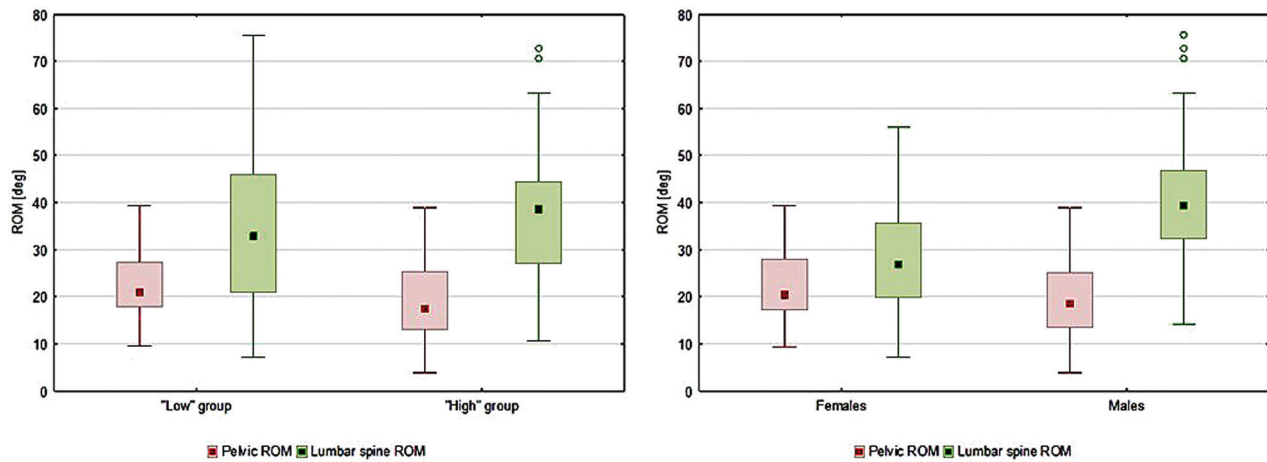


Figure 3. Pelvic and lumbar spine ranges of motion during squatting. Boxes show median, upper and lower quartiles. Whiskers show maximum and minimum (1.5 of interquartile range). Dots show outliers (above 1.5 of interquartile range). None of the comparisons between sitting groups are statistically significant. There is a statistical difference in lumbar ROM between sexes ($p < 0.01$).

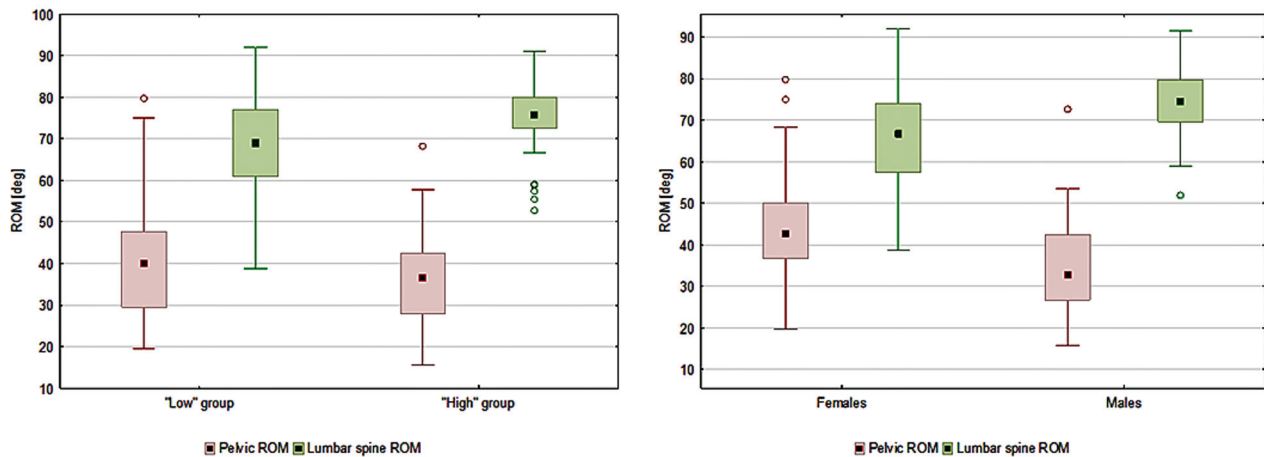


Figure 4. Pelvic and lumbar spine ranges of motion during forward bending. Boxes show median, upper and lower quartiles. Whiskers show maximum and minimum (1.5 of interquartile range). Dots show outliers (above 1.5 of interquartile range). There is a statistically significant difference in lumbar spine ROM between groups ($p = 0.02$). All comparisons between sexes are statistically significant ($p < 0.05$).

($Me = 74.52$ vs $Me = 66.67$; $Z = -2.46$, $p = 0.01$, $ES = 0.38$) but smaller pelvic ROM ($Me = 32.76$ vs $Me = 42.73$; $Z = 2.89$, $p < 0.01$, $ES = 0.44$) than females did.

3.2. Angular velocities and EMG

Statistical analysis of angular velocities revealed that average ($Me = 20.60$ vs $Me = 16.99$) and peak velocities ($Me = 50.07$ vs $Me = 40.56$) of lumbar spine during squatting ascent were significantly greater in the 'high' than in the 'low' sitting group ($p = 0.4$, $ES = 0.30$ in both cases). There were no statistically significant differences between groups in velocities of the pelvis and of the spine during forward bending. Details of statistical analysis are presented in Tables 3 and 4.

Males showed significantly greater average velocities of lumbar spine during the ascent phases of squatting and forward bending comparing with females ($Me = 20.23$ vs $Me = 16.42$, $Z = -2.29$, $p = 0.02$, $ES = 0.35$; $Me = 35.69$ vs $Me = 28.72$, $Z = -3.39$, $p < 0.001$, $ES = 0.52$, respectively). Females demonstrated greater average velocities of pelvis than males during forward bending descent ($Me = 14.57$ vs $Me = 10.7$, $Z = 2.19$, $p = 0.03$, $ES = 0.33$). Males also showed greater peak velocities of lumbar spine during squatting descent and ascent and during forward bending ascent ($Me = 34.06$ vs $Me = 24.38$, $Z = -2.43$, $p = 0.01$, $ES = 0.37$; $Me = 50.30$ vs $Me = 35.12$, $Z = -3.03$, $p < 0.01$, $ES = 0.46$; $Me = 75.77$ vs $Me = 60.19$, $Z = -2.75$, $p = 0.01$, $ES = 0.42$, respectively).

Table 3. Average angular velocities during squatting and forward bending.

Average angular velocities [deg/s]	'Low' sitting time			'High' sitting time			Statistics		
	M	Me	SD	M	Me	SD	U	Z	p
Squatting descent									
Pelvis	13.06	11.33	5.75	11.00	11.06	5.57	392.00	1.39	0.17
Lumbar spine	14.72	11.10	7.85	16.36	15.08	6.56	366.00	-1.74	0.08
Squatting ascent									
Pelvis	12.45	11.37	4.95	10.79	9.42	4.90	384.00	1.50	0.13
Lumbar spine	18.71	16.99	8.93	21.72	20.60	7.28	345.00	-2.03	0.04
Forward bending descent									
Pelvis	12.85	13.00	5.68	12.23	11.86	4.97	473.00	0.27	0.79
Lumbar spine	21.29	21.18	6.00	24.91	22.86	8.75	375.00	-1.62	0.11
Forward bending ascent									
Pelvis	17.98	17.22	6.71	15.76	15.45	5.19	394.00	1.36	0.17
Lumbar spine	31.71	30.78	6.44	35.76	34.77	10.02	375.00	-1.62	0.11
Average angular velocities [deg/s]	Females			Males			Statistics		
	M	Me	SD	M	Me	SD	U	Z	p
Squatting descent									
Pelvis	13.42	11.30	5.80	11.36	11.19	5.61	375.00	1.21	0.23
Lumbar spine	13.11	12.56	5.18	16.84	13.93	7.98	353.00	-1.52	0.13
Squatting ascent									
Pelvis	12.79	11.75	4.77	11.06	9.82	5.01	351.00	1.55	0.12
Lumbar spine	16.35	16.42	6.94	22.26	20.23	8.31	299.00	-2.29	0.02
Forward bending descent									
Pelvis	14.79	14.57	6.06	11.28	10.71	4.45	306.00	2.19	0.03
Lumbar spine	21.27	21.53	5.70	23.93	23.12	8.35	384.00	-1.08	0.28
Forward bending ascent									
Pelvis	19.15	17.37	7.11	15.70	16.04	5.14	331.00	1.83	0.07
Lumbar spine	29.23	28.72	5.46	36.08	35.69	8.91	222.00	-3.39	<0.001

M: Mean; Me: median; SD: standard deviation.

Table 4. Peak angular velocities during squatting and forward bending.

Peak angular velocities [deg/s]	'Low' sitting time			'High' sitting time			Statistics		
	M	Me	SD	M	Me	SD	U	Z	p
Squatting descent									
Pelvis	30.18	26.48	12.76	27.10	24.56	13.68	422.00	0.97	0.33
Lumbar spine	34.45	25.88	18.93	36.18	33.83	17.31	411.00	-1.12	0.26
Squatting ascent									
Pelvis	28.11	25.21	11.81	25.97	23.24	13.34	407.00	1.18	0.24
Lumbar spine	40.47	40.56	17.90	52.29	50.07	27.09	344.00	-2.05	0.04
Forward bending descent									
Pelvis	28.66	28.31	13.85	30.33	31.36	12.42	428.00	-0.89	0.37
Lumbar spine	47.83	47.49	17.87	56.55	52.45	22.39	381.00	-1.54	0.12
Forward bending ascent									
Pelvis	38.89	39.82	13.07	35.47	33.58	11.34	400.00	1.28	0.20
Lumbar spine	64.23	64.10	15.34	76.41	71.66	27.18	360.00	-1.83	0.07
Peak angular velocities [deg/s]	Females			Males			Statistics		
	M	Me	SD	M	Me	SD	U	Z	p
Squatting descent									
Pelvis	31.29	26.74	12.52	27.31	24.34	13.47	362.00	1.39	0.17
Lumbar spine	29.25	24.38	15.94	38.70	34.06	18.51	289.00	-2.43	0.01
Squatting ascent									
Pelvis	27.19	24.33	9.66	27.09	23.25	13.96	397.00	0.89	0.38
Lumbar spine	35.29	35.12	15.39	52.01	50.30	24.82	247.00	-3.03	<0.01
Forward bending descent									
Pelvis	32.74	30.87	14.69	27.53	25.47	11.92	381.00	1.12	0.26
Lumbar spine	47.27	46.82	16.07	54.48	54.22	22.25	369.00	-1.29	0.20
Forward bending ascent									
Pelvis	40.88	38.77	13.34	35.27	34.25	11.38	349.00	1.58	0.12
Lumbar spine	60.27	60.19	13.91	75.33	75.77	24.39	267.00	-2.75	0.01

M: Mean; Me: median; SD: standard deviation.

There were no significant differences in FRR and ERR between groups during squatting (FRR right: Me = 2.04 vs Me = 1.87, FRR left: Me = 1.76 vs Me = 2.10, ERR right: Me = 1.94 vs Me = 1.85, ERR left: Me = 1.93 vs

Me = 2.05) and forward bending (FRR right: Me = 3.28 vs Me = 7.20, FRR left: Me = 3.44 vs Me = 5.34, ERR right: Me = 5.66 vs Me = 8.62, ERR left: Me = 5.67 vs Me = 7.64). However, differences in FRR of the left

Table 5. Multifidus muscles flexion-relaxation phenomenon during squatting and forward bending.

Task	Multifidus muscle	'Low' sitting time			'High' sitting time			Statistics		
		M	Me	SD	M	Me	SD	U	Z	p
Squatting	FRR right	4.30	2.04	6.29	5.92	1.87	11.83	478.00	0.20	0.84
	ERR right	3.32	1.94	4.01	3.82	1.85	5.07	487.00	−0.08	0.94
	FRR left	2.42	1.76	1.91	7.97	2.10	14.52	424.00	−0.94	0.34
	ERR left	2.58	1.93	1.76	3.55	2.05	3.99	491.00	0.02	0.98
Forward bending	FRR right	10.70	3.82	14.71	13.55	7.20	23.01	459.00	−0.46	0.64
	ERR right	9.55	5.66	13.26	15.38	8.62	26.57	421.00	−0.99	0.32
	FRR left	7.78	3.44	9.89	18.01	5.34	30.99	351.00	−1.95	0.051
	ERR left	8.31	5.67	9.67	12.98	7.64	15.48	403.00	−1.23	0.22

Task	Multifidus muscle	Females			Males			Statistics		
		M	Me	SD	M	Me	SD	U	Z	p
Squatting	FRR right	6.65	1.86	11.72	4.12	2.01	7.42	398.00	0.88	0.38
	ERR right	3.71	1.73	4.52	3.45	2.03	4.54	441.00	−0.26	0.79
	FRR left	3.22	1.67	5.31	5.98	2.20	12.17	362.00	−1.39	0.17
	ERR left	2.23	1.75	1.69	3.48	2.43	3.50	346.00	−1.62	0.11
Forward bending	FRR right	16.62	4.47	28.14	9.36	4.58	10.00	445.00	−0.21	0.84
	ERR right	16.35	5.52	31.55	9.87	7.36	9.59	404.00	−0.79	0.43
	FRR left	10.30	3.21	15.51	13.75	4.87	25.97	400.00	−0.85	0.40
	ERR left	9.47	4.45	14.24	11.03	7.68	12.02	375.00	−1.21	0.23

FRR: Flexion-relaxation ratio; ERR: extension-relaxation ratio; M: mean; Me: median; SD: standard deviation.

multifidus during forward bending were greater in the 'high' than in the 'low' sitting group (Me = 5.34 vs Me = 3.44; $Z = -1.95$, $p = 0.051$). This result is close to the accepted level of significance. There were no significant differences in FRR and ERR between females and males during squatting (FRR right: Me = 1.86 vs Me = 2.01, FRR left: Me = 1.67 vs Me = 2.20, ERR right: Me = 1.73 vs Me = 2.03, ERR left: Me = 1.75 vs Me = 2.43) and forward bending (FRR right: Me = 4.47 vs Me = 4.58, FRR left: Me = 3.21 vs Me = 4.87, ERR right: Me = 5.52 vs Me = 7.36, ERR left: Me = 4.45 vs Me = 7.68) (Table 5).

4. Discussion

The aim of this study was to investigate lumbar-pelvic kinematics and MF activity during squatting and forward bending in relation to the sedentary behaviour of physically active young adults. The results of this study are generally in agreement with the stated hypothesis. Increased time in sitting significantly alters lumbar-pelvic kinematics during squatting and forward bending compared to less sitting time. This conclusion can be applied to people who meet PA guidelines.

4.1. Lumbar-pelvic ratios and ROMs

Greater values of the lumbar-pelvic ratio in the 'high' sitting group indicates that the domination of lumbar spine motion over pelvic motion during tasks is increased as a result of prolonged sitting. Previous research links greater lumbar-pelvic or lumbar-hip ratios with sex and age. The lumbar contribution to movement has been identified as smaller in females

than in males and smaller in older than in younger participants (Vazirian et al. 2017; Pries et al. 2015). However, in this study groups were balanced with respect to sex and age. Patients suffering from LBP also have been reported to have decreased lumbar motion in forward bending tasks (Larivière, Gagnon, and Loisel 2000). On the other hand, Jandre Reis and Macedo (2015) have shown that lumbar mobility was greater while at the same time pelvic mobility was restricted in LBP patients comparing with controls. Lumbar-pelvic movement patterns in LBP patients still remain unclear. It may be that people with LBP should be classified into subgroups by the pain provoked by movement (flexion/extension), As was suggested by Kim et al. (2013) and Laird et al. (2014) lower movement speed is commonly seen in people with LBP and has been linked to a fear of movement (Thomas et al. 2008).

We consider two possible explanations of our current results. First, thigh hamstrings can restrict pelvis anterior rotation and lead to compensation resulting in increased lumbar flexion. Second, deconditioning of the lumbar muscles and excessive loading of passive tissue in a sitting position can lead to changes in the trunk movement pattern. Previous studies of Hasebe et al. (2014) suggest that subjects with tight hamstrings have decreased pelvic mobility. A compensatory increase in the lumbar ROM during the forward bending movement can be the result of a slightly, but not statistically significant decreased pelvic mobility in the 'high' sitting group. In clinical practice, limitation of hamstring flexibility and greater lumbar spine ROM should be an indicative of low back stabilisation exercises and hamstring

stretching (Jandre Reis and Macedo 2015). The dynamic stretching of hamstrings significantly improves maximal forward bending and pelvic mobility, which decreased the lumbar-pelvic ratio (Hasebe et al. 2016). Long-duration sitting can be a contributory factor in hamstring tightness (Waqas et al. 2016; Fatima et al. 2017). On the other hand, continuous sitting also resulted in higher trunk flexion angles (Jia and Nussbaum 2018). Sustained trunk flexion during sitting can lead to creep of back viscoelastic tissues which can increase the risk of LBP (Little and Khalsa 2005). Creep of lower back extensor muscles may occur due to prolonged passive stretching in a sitting position. That may lead to a reduced contribution of passive tissue in maintaining spinal stability and posture. Thus, as a result of compensation, lower back extensor muscles are required to generate more active forces during static and dynamic tasks and cause fatigue.

4.2. Angular velocities and EMG

The reduced ability of muscles to generate strength may be a significant risk factor for low back pain (G. Shin, D'Souza, and Liu 2009). P. O'Sullivan et al. (2006) observed decreased activity in lumbar MF while moving from upright sitting to slump sitting, suggesting the occurrence of relaxation of the spinal stabilising muscles. However, the muscle activity pattern was highly variable. In our study there were no statistically significant differences between groups. However, FRR of the left multifidus during forward bending was greater in the 'high' than in the 'low' sitting group. That finding may be the result of greater lumbar spine ROM during forward bending in the 'high' sitting group. However, the sEMG results are rather inconclusive which is in line with O'Sullivan's previous study. With regard to the large data scatter our results of sEMG should be interpreted with caution.

Greater angular velocity of the spine observed in the 'high' sitting group could be considered a risk factor in back muscles injuries. Yoon, Shiekhzadeh, and Nordin (2012) suggests that faster movement during loading tasks significantly affects passive spinal elements due to the viscoelastic characteristics of tissues. Greater acceleration of movement requires coordinated muscle recruitment to initiate and continue the movement.

The different FRP levels found in participants with different flexibilities can be explained by the change of lumbar curvature. Previously, it has been demonstrated that participants with low flexibility exhibited more evident FRP (Chen et al. 2018). Contrary, we

have noticed that high sitting group, which showed greater lumbar-pelvic ratio, demonstrated greater FRP, nevertheless this finding was not statistically significant and needs further investigations.

4.3. Limitations and future research

Our study has a few limitations. First, the effect sizes of statistically significant comparisons between groups are rather small (0.30–0.37). Thus, current results should be interpreted with caution and need further investigation.

Second, we did not investigate hamstring extensibility and we consider this a main limitation of our study. Previously, angular tests measured by goniometers, such as active knee extensions, have been recommended as criterion measures (Mayorga-Vega, Merino-Marban, and Viciano 2014). An analysis of the relationship between hamstring flexibility, sitting time, and lumbar-pelvic kinematics may justify our reasoning about the causes of the current results.

According to the authors' knowledge, this is the first study investigating the relationship between sitting behaviour and lumbar-pelvic kinematics during squatting and forward bending. Due to the fact that we have investigated only physically active people, it should be worth investigating whether the association between sitting time and lumbar-pelvic kinematics is stronger in non-physically active persons or office workers.

4.4. Practical applications

This study has practical implications for students and other young people continuing their education who spend a lot of time sitting but are trying to meet the PA recommendation. Our study shows that lumbar pelvic kinematics can be affected by sitting time even when physical activity level is sufficient or high. Therefore, sedentary behaviour should be taken into account by physical therapists, coaches and other professionals when evaluating squats and forward flexion movements. Higher education is a critical period of life when health behaviours can be formed in young adults. Thus, education for reducing sedentary behaviours among students could be a public health strategy for LBP prevention (Vainshelboim et al. 2019).

According to Heneghan et al. (2018) it is probable that young adults are a 'risk' group for developing future musculoskeletal complaints like LBP, and it is very likely that they will work in occupations where substantial periods of time will be sitting.

Understanding lumbar-pelvic kinematics can provide new information about not clear relationship of sedentary behaviour and risk of LBP. Several biomechanical studies have been conducted to clarify the relationship between sitting and LBP (Torén 2001; Mork and Westgaard 2009). Our investigation can be considered as a starting point for making more specific research about how sitting time is associated with biomechanics during functional dynamic tasks to those accrued through engaging in moderate-to-vigorous physical activity.

5. Conclusions

Lumbar-pelvic kinematics are significantly altered by longer sitting times in physically active people. Lumbar-pelvic ratios during squatting and forward bending and spine velocity during squatting were greater in participants who have been sitting longer than 7 h/day. That suggests that sitting time can be considered one of the factors influencing lumbar-pelvic biomechanics during dynamic tasks. Thus, it should be taken into account during an assessment of squatting and forward bending movements. Moreover, our study extends our understanding of sitting behaviours and tries to explain the relationship between longer sitting time and potential LBP risk.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

The author(s) reported there is no funding associated with the work featured in this article.

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Data availability statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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